Computational Assessment of Oxygen Delivery in Decellularized Leaf Vascular Scaffolds

Pipeline Development, GreensV4 Failure Analysis, and a Graph-Based Oxygen Screening Prototype

Capstone manuscript

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Contents

Abstract **3**

1 Introduction **3**

2 Materials and Methods **4**

2.1 Computational workflow 4

2.2 Image data source and LeafVeinCNN preprocessing 5

2.3 MAT-to-NetFlowV2 graph export and network preparation 7

2.4 Network hemodynamics with NetFlowV2 9

2.5 GreensV4 testing and failure-diagnosis procedure 9

2.6 GreensV4 failed by memory scaling on full-resolution plant networks . . . 10

2.7 GreensV4 failed by directed convective-ordering incompatibility 10

2.8 Parameter tuning could not repair the GreensV4 topology failure 11

3 Proposed OxySim oxygen-screening solver **11**

3.1 Inputs and parameter control 12

3.2 Governing tissue oxygen equation 12

3.3 Radius-aware wall exchange 13

3.4 Internal skeleton hemodynamics 13

3.5 Lumen oxygen transport and haemoglobin option 14

3.6 Coupled simulation loop 14

3.7 Exactly what the solver calculates 15

3.8 Batch sweep execution 16

3.9 Limitations of the proposed solver 17

3.10 Equation justification and source discipline 18

3.11 Parameterization and sweep design 21

3.12 Verification boundary and R Markdown stability analysis 21

4 Results **22**

4.1 Image-to-flow conversion was computationally feasible 22

5 Discussion **22**

5.1 Scientific meaning of the negative result 22

5.2 Why NetFlowV2 remains useful 23

5.3 Status of the proposed oxygen-screening solver 23

6 Limitations **23**

7 Conclusion **24**

References **24**

Abstract

Background: Decellularized plant scaffolds provide naturally branched and reticulated vascular architectures that may be useful for tissue engineering, but their ability to support oxygen delivery to seeded mammalian cells remains computationally and experimentally uncertain. **Objective:** This study developed an image-driven computational workflow for converting leaf venation data into perfusion-resolved vascular graphs and evaluated whether an established microvascular Green's-function oxygen solver could be transferred directly to reticulated plant networks. **Methods:** LeafVeinCNN-derived segmentations were converted into NetFlowV2-compatible vascular graphs with calibrated geometry, radius estimates, and reproducible boundary-condition annotations. NetFlowV2 was used to solve steady graph hemodynamics. GreensV4 was then tested as an oxygen-transport solver, and failure modes were traced through memory scaling, flow-direction ranking, and intravascular convective dependency logic. A graph-based oxygen-screening prototype was implemented as an alternative route that solves dissolved intraluminal oxygen simultaneously on the directed flow graph and rasterizes the result for tissue accessibility metrics. **Results:** NetFlowV2 remained suitable for solving pressure-flow fields on reticulated networks. GreensV4 was not directly suitable for full-resolution plant venation because dense interaction matrices scaled quadratically with the number of vessel source points and because its convective oxygen routine required an upstream-to-downstream ordering that can be invalidated by non-rankable convective dependencies. The graph-based prototype generated executable oxygen-accessibility outputs, including vessel oxygen maps, tissue screening maps, distance-to-vessel fields, and scalar threshold metrics. **Conclusion:** The main contribution is not a validated biological prediction of scaffold viability; it is a reproducible computational pipeline, a documented algorithmic failure analysis, and a preliminary topology-compatible oxygen-screening method that can be validated in future work.

Keywords: decellularized plant scaffold; leaf venation; NetFlowV2; GreensV4; oxygen transport; graph solver; tissue engineering; computational failure analysis

1. Introduction

Decellularized plant tissues have been proposed as inexpensive, structurally complex scaffolds for tissue engineering because plant vascular networks provide natural branching, hierarchical transport pathways, and large internal surface area. Several studies have demonstrated that plant-derived scaffolds can be decellularized, perfused, and reseeded with mammalian cells. However, structural preservation alone does not establish whether a scaffold can maintain oxygen delivery under realistic culture conditions.

Oxygen transport is a central constraint in engineered tissues. Mammalian cells consume oxygen continuously, and insufficient delivery produces hypoxia, growth arrest, or cell

death depending on local partial pressure and exposure duration. Therefore, a useful computational scaffold-screening workflow must connect image-derived vascular geometry to perfusion, diffusion, and consumption metrics rather than reporting geometry alone.

This project began with the objective of building an end-to-end oxygenation pipeline for decellularized leaf scaffolds. The original plan was to use LeafVeinCNN for image segmentation, NetFlowV2 for hemodynamics, and GreensV4 for Green's-function oxygen transport. The final project was reframed after systematic testing showed that GreensV4 could not be directly transferred to full reticulated plant venation without violating important numerical and topological assumptions. This negative result became a central methodological finding rather than a defect.

The revised objective was therefore to document the image-to-flow pipeline, explain precisely why GreensV4 failed, and implement a preliminary oxygen-screening method that does not require a tree-like vascular ordering. The manuscript is written with strict claim discipline: the graph oxygen output is treated as a screening prototype, not as a validated substitute for experimental measurements or for a fully verified three-dimensional transport solver.

The specific contributions are as follows:

1. Construction of a reproducible LeafVeinCNN-to-NetFlowV2 pipeline for converting image-derived plant venation into solver-ready hemodynamic graphs.
2. Computational failure analysis showing that GreensV4 is not directly compatible with full-resolution reticulated plant networks because of dense-matrix memory scaling and invalid directed convective ordering.
3. Implementation of a proposed OxySim oxygen-screening prototype that reconstructs pixel-skeleton hemodynamics from LeafVeinCNN MAT exports, applies radius-aware wall exchange, advances a two-dimensional reaction-diffusion tissue field, and outputs oxygen-threshold screening metrics rather than experimentally validated viability predictions.
4. Generation of demonstration outputs suitable for a capstone-level results section: NetFlow mass-balance metrics, failure-mode documentation, vessel oxygen maps, tissue oxygen-accessibility maps, and distance-to-vessel metrics.

2. Materials and Methods

2.1. Computational workflow

The workflow was organized into five stages: image segmentation, graph reconstruction, network hemodynamics, and oxygen-screening analysis. The oxygen stage was modified

after GreensV4 failure analysis, but the upstream segmentation and hemodynamic stages remained valid components of the pipeline.

Table 1. Computational workflow used in the reframed manuscript.

Stage	**Purpose**	**Input**	**Output**
LeafVeinCNN segmentation	Detect full-width venation, skeletonize the vascular network, and estimate vessel width.	Calibrated leaf image or exported MATLAB file.	Vessel mask, skeleton, edge list, width/radius estimates, and quality-control overlays.
Graph preparation	Convert the skeleton into a solver-ready vascular graph and enforce boundary-condition consistency.	Segmentation outputs and pixel-to-micrometre calibration.	NetFlowV2 Network.dat with nodes, segments, diameters, boundary nodes, and parameter files.
NetFlowV2 hemodynamics	Solve pressure and flow on the vascular graph using Poiseuille/Kirchhoff conservation.	Network.dat, RheolParams.dat, ContourParams.dat.	NetworkNew.dat with segment flows, pressures, and hemodynamic diagnostics.
GreensV4 failure analysis	Test whether the reference Green's-function oxygen solver can process the reticulated plant graph.	NetworkNew.dat and oxygen parameter files.	Failure logs, memory-scaling analysis, and topological-ordering diagnosis.
Proposed OxySim oxygen-screening prototype	Generate preliminary oxygen maps and threshold metrics after GreensV4 failure without claiming experimental validation.	LeafVeinCNN MAT geometry and JSON parameter file.	Tissue O2 arrays, lumen O2 arrays, pO2 maps, distance-to-vessel metrics, threshold area fractions, and batch summaries.

2.2. Image data source and LeafVeinCNN preprocessing

The image-processing stage used cleared leaf venation images from the Zenodo CHAMBASA leaf venation image dataset. The Zenodo record describes images collected from forest plots in Peru for the CHAMBASA plant trait campaign and notes that the images were produced after chemical leaf clearing. The processed archive also includes contrast-enhanced images with the suffix '-CLAHE'. These images were selected because cleared-leaf preparations expose venation architecture more clearly than untreated leaf photographs and are therefore more appropriate for skeletonization, vein-width estimation, and downstream graph extraction. In the present project, the CHAMBASA images were not treated as biological perfusion experiments; they were used as calibrated geometric inputs for a computational scaffold model.

The selected images were processed in LeafVeinCNN using the dataset-provided spatial calibration of 9.62 μm/px. This calibration was entered during LeafVeinCNN preprocessing and retained in the exported MATLAB files through the MicronPerPixel field. The same calibration was then propagated into all downstream metric conversions, including segment length, vessel width, vessel diameter, node coordinates, and tissue-distance calculations. Calibration consistency is critical because all downstream physical quantities are derived from image-space coordinates; changing the pixel-to-micrometre conversion would rescale vessel lengths, radii, diffusion distances, and hydraulic resistance.

All images were rescaled in LeafVeinCNN using a rescale factor of 2 before CNN prediction. This was done to improve the representation of fine veins relative to the receptive scale of the trained models. Rescaling was not interpreted as adding new biological resolution; it increased interpolated pixel sampling for CNN segmentation while the physical interpretation remained controlled by the fixed 9.62 μm/px calibration supplied in the data-processing workflow.

CNN prediction was run in random mode with an ensemble size of 3. In this setting, three trained CNN models were selected randomly and their prediction maps were averaged before segmentation. The reason for using a small random ensemble was to reduce dependence on the idiosyncrasies of a single trained model. This does not eliminate all model bias, but it lowers single-network selection bias and gives a more stable venation probability map than relying on one CNN alone.

The outputs retained from LeafVeinCNN for downstream modelling were the full-width binary vein mask, the one-pixel skeleton, centreline edge information, graph-related connectivity data, and calibrated width estimates. The full-width mask was used for vessel diameter estimation and for excluding vessel pixels from tissue-domain calculations. The skeleton and centreline edge list were used to define the topological graph. This separation is necessary because the full-width mask defines vessel area, whereas the skeleton defines graph connectivity.

Processing settings summary for the LeafVeinCNN stage.

Processing item	**Value used**	**Methodological reason**
Image source	Zenodo CHAMBASA cleared leaf venation image dataset	Provides cleared venation images suitable for graph extraction.
LeafVeinCNN calibration	9.62 μm/px	Converts image coordinates, widths, lengths, and distance maps into physical units.
Image rescaling	2x	Improves CNN sampling of fine venation; it does not create new biological resolution.

Processing item	Value used	Methodological reason
CNN mode	Random	Avoids manually selecting a single preferred CNN model.
CNN ensemble size	3	Averages multiple predictions to reduce single-model dependence.
Primary retained outputs	Full-width mask, skeleton, centreline edge list, width estimates, MATLAB export	Supplies both geometric vessel area and graph topology for NetFlowV2 export.

2.3. MAT-to-NetFlowV2 graph export and network preparation

The MATLAB export from LeafVeinCNN was converted into a NetFlowV2-compatible vascular graph using two Python scripts: leafveincnn_to_netflow.py and prepare_network.py. The first script performed the direct MAT-to-Network.dat conversion. The second script acted as a cleaning and validation wrapper that called the exporter, corrected numerical pathologies, validated boundary conditions, and wrote the final network_ready.dat file used for NetFlowV2.

The exporter treated the LeafVeinCNN centreline edge list as the graph source. Each centreline polyline was converted from image coordinates into physical coordinates using the pixel calibration. The coordinate convention was row-column image space converted to x-y coordinates in micrometres, with z set to zero because the scaffold image represented a planar cleared-leaf preparation. Edges were then resampled into NetFlowV2 segments so that curved veins could be represented by a sequence of short straight segments rather than by a single chord. This is important because NetFlowV2 computes segment length from node coordinates, and a long curved edge represented as one straight segment would underestimate geometric length and therefore overestimate conductance.

Segment diameters were assigned from LeafVeinCNN widthCalibrated values when available. When calibrated width values were absent or unusable, the fallback diameter estimate was obtained from twice the Euclidean distance transform of the full-width vessel mask. A minimum diameter floor was applied to prevent zero-diameter segments. This radius-handling step was treated as a major uncertainty channel because Poiseuille conductance scales approximately with diameter to the fourth power. Consequently, diameter quality control is essential; it directly determines the magnitude and stability of the hemodynamic solution.

Boundary conditions were assigned only after export-graph discretization. This was essential because NetFlowV2 requires boundary nodes to be terminal, degree-1 nodes in the segment graph. If a pressure boundary condition is placed on an interior bifurcation, the solver can report boundary-node errors and the pressure system can become ill-defined.

The exporter therefore validated boundary nodes against the exported segment graph rather than only against the raw LeafVeinCNN edge graph. Degree-zero orphan nodes were removed from the boundary list, and interior boundary nodes were replaced by the nearest degree-1 node when a valid terminal could be reached. This prevented a recurrent failure mode in which visually plausible boundary choices became invalid after discretization.

The network preparation wrapper then applied a second validation layer. It capped very large diameters, removed junction artefacts shorter than 0.5 μm, identified disconnected components, removed small isolated fragments, and preserved large disconnected boundary fragments as unperfused rather than forcing them into the main hydraulic network. Near-touching components separated by sub-pixel gaps could be merged when the gap was consistent with rounding or endpoint-snapping artefacts. The final pass confirmed that every boundary-condition node was degree-1 before writing the final network file.

The final graph-preparation outputs were Network.dat, network_ready.dat, export_report.txt or export_report.json when available, and diagnostic logs describing cap operations, short-segment removal, disconnected components, boundary-condition repair, and final validation.

Computational operations used to convert LeafVeinCNN MAT exports into NetFlowV2 inputs.

Operation	**Implementation**	**Purpose**
MAT loading	Read bwFullWidth, skeleton, widthCalibrated, edgelistCenter, and MicronPerPixel when present.	Recover vessel area, topology, widths, and calibration from LeafVeinCNN.
Polyline conversion	Convert centreline paths from row-col image coordinates to x-y micrometre coordinates.	Place the plant venation graph into the physical coordinate system required by NetFlowV2.
Edge resampling	Represent curved veins as chains of short straight NetFlowV2 segments.	Preserve arc length more accurately than single straight-edge approximations.
Diameter assignment	Use widthCalibrated primarily; use 2x distance-transform diameter as fallback.	Attach transport-relevant hydraulic geometry to each segment.
Boundary-condition validation	Force all BC nodes to be degree-1 terminals after discretization.	Satisfy the NetFlowV2 terminal boundary-condition requirement.

Operation	Implementation	Purpose
Post-export repair	Cap extreme diameters, remove sub-0.5 μm artefacts, classify disconnected components, and revalidate BCs.	Improve numerical conditioning and prevent hidden topology errors before NetFlowV2.
Final outputs	network_ready.dat, Network.dat, reports, and repair logs.	Provide reproducible NetFlowV2-ready inputs and auditable preprocessing history.

2.4. Network hemodynamics with NetFlowV2

NetFlowV2 was used to solve the pressure-flow problem on the extracted vascular graph. This stage is mathematically distinct from oxygen transport: NetFlowV2 solves a global linear network problem in which pressure differences drive segment flows and interior nodes satisfy conservation of volumetric flow.

Reticulated loops do not invalidate the NetFlowV2 pressure-flow solve. Loops are handled through the global Kirchhoff system, provided that boundary conditions are specified and the graph is connected in a physically interpretable way. This distinction is important because GreensV4 failed later at the intravascular oxygen-convection stage, not at the hemodynamic stage.

$$Q_e = G_e(P_i - P_j), \qquad G_e \propto \frac{r_e^4}{\mu L_e}. \tag{1}$$

Here Q_e is the segment flow rate, G_e is hydraulic conductance, P_i and P_j are endpoint pressures, r_e is segment radius, μ is viscosity, and L_e is segment length.

2.5. GreensV4 testing and failure-diagnosis procedure

GreensV4 was tested as the downstream oxygen solver because Green's-function methods are established for microvascular oxygen transport. However, the solver was developed for vascular geometries whose intravascular transport can be processed through a valid downstream ordering. Full plant venation violates this assumption in common cases because leaf networks are reticulated and contain closed loops.

Two failure classes were analyzed. The first was a memory-scaling failure caused by dense vessel-vessel and solver matrices. The second was a topological-ordering failure caused by reticulated topology, reversed segment orientation, near-zero-flow edges, and multiple upstream dependencies after NetFlowV2 assigned flow directions. These failures

were separated because memory can sometimes be mitigated by coarsening, whereas non-rankable convective dependencies require a different transport algorithm.

2.6. GreensV4 failed by memory scaling on full-resolution plant networks

The first GreensV4 failure was an allocation failure during matrix construction. In the representative case, the solver reported approximately 61,021 vessel source points and 98,368 tissue points before failing during matrix allocation. This is consistent with dense matrix scaling: a single float matrix of size 61,021 × 61,021 requires approximately 14.9 GB before overhead. Multiple dense matrices of comparable size exceed ordinary workstation memory.

2.7. GreensV4 failed by directed convective-ordering incompatibility

The second failure was algorithmic. GreensV4 propagates intravascular oxygen downstream through the flow graph. This requires a valid order in which every node can be processed after all of its upstream inflow segments have already been processed. Reticulated leaf venation can violate this condition after flow directions are assigned because the solver dependency structure can become non-rankable in loops or near-zero-flow regions.

In a non-rankable convective dependency structure, no node can be processed first without requiring information from a segment that has not yet been processed. NetFlowV2 can solve pressure and flow on reticulated networks because it solves a global linear system. GreensV4's convective oxygen pass is stricter because it recursively propagates oxygen concentrations and sensitivity coefficients through a one-way processing sequence.

The observed failure signature was consistent with this mechanism: NetFlowV2 completed the hemodynamic solve, but GreensV4 generated convective sequence errors, flow-conservation warnings, invalid oxygen concentrations, or undefined outputs. The failure therefore occurred after the hemodynamic stage, during intravascular oxygen transport.

Interpretation of the GreensV4 failure.

Observation	**Interpretation**	**Consequence**
Large vessel-source count followed by allocation failure.	Dense interaction matrices scale as $O(nv^2)$.	Full-resolution plant networks exceed feasible memory.
Wrong segment sequence during convection.	At least one inflow segment was required before being processed.	The directed graph was not rankable as a one-way downstream tree.

Observation	Interpretation	Consequence
Flow-conservation warnings at oxygen stage.	The oxygen-convection routine received inconsistent or incomplete upstream data.	Invalid node oxygen values can propagate through the solver.
NaN or empty oxygen outputs.	The transport iteration became numerically undefined.	GreensV4 outputs cannot be used as biological evidence.

The ordering failure has a biological basis. Leaf venation is not a simple tree. Reticulate leaf networks contain closed loops that improve redundancy and robustness to damage or fluctuating local demand. Those loops are structurally relevant for the plant scaffold concept because they are precisely the architectural feature that makes leaf tissue attractive as a naturally branched transport template (Katifori et al., 2010; Sack and Scoffoni, 2013; Katifori, 2018).

Therefore, pruning loops only to satisfy the GreensV4 convective order would solve the implementation-level failure while destroying the biological topology under investigation. A tree-like approximation could be used as a simplified control, but it cannot be presented as the full reticulated scaffold. This is the central reason that the GreensV4 failure is a domain-of-validity failure rather than a minor implementation defect.

2.8. Parameter tuning could not repair the GreensV4 topology failure

Changing biological oxygen parameters cannot produce a valid upstream ordering when the directed graph contains a dependency cycle. Parameters such as wall permeability, tissue diffusivity, oxygen consumption, P50, Hill coefficient, inlet oxygen, and effective exchange coefficients only affect oxygen transport after the intravascular oxygen field has a valid computational structure.

Similarly, aggressive low-flow pruning is not an acceptable primary solution. If weakly perfused loop segments are removed only to force a rankable graph, the model no longer represents the reticulated leaf scaffold. It becomes an artificial tree extracted from a leaf network. That would answer a different biological question.

3. Proposed OxySim oxygen-screening solver

After GreensV4 failed on the full-resolution reticulated plant networks, a proposed oxygen-screening solver was implemented in oxy_sim_mat_saveO2.py. The purpose of this solver was not to replace experimental validation or to claim final physiological accuracy. Its purpose was to provide a controllable computational route for estimating dissolved oxygen accessibility from LeafVeinCNN-derived plant scaffold geometry while explicitly recording the numerical assumptions, stabilizing safeguards, and output metrics.

The solver is best described as a coupled pixel-skeleton hemodynamics and two-dimensional oxygen-screening model. It reads a LeafVeinCNN MATLAB export directly, reconstructs a vessel mask and skeleton on a simulation grid, estimates radius-dependent wall exchange, solves a graph Poiseuille pressure-flow problem on the skeleton pixels, advects intraluminal oxygen along the skeleton, and updates tissue oxygen concentration by explicit reaction-diffusion with Michaelis-Menten consumption. Therefore, this code is a proposed oxygen-screening layer. It is not described as a fully validated NetFlowV2 downstream oxygen solver, because the hemodynamic field is reconstructed internally on the resized skeleton rather than read directly from NetFlowV2 NetworkNew.dat.

3.1. Inputs and parameter control

The solver is controlled by a strict JSON configuration file. The configuration defines the input MAT file, output directory, grid resolution, simulation duration, oxygen diffusivity, oxygen consumption parameters, wall permeability, radius scaling, exchange clamp, inlet oxygen, hemodynamic pressure boundary values, fluid viscosity, velocity floor, optional haemoglobin parameters, convergence controls, and distance-bin settings. This design improves reproducibility because every run is reproducible from one saved configuration file rather than from hidden interactive choices.

In the MAT-direct mode used here, the solver reads the LeafVeinCNN fields bwFullWidth, skeleton, widthCalibrated when available, and MicronPerPixel. The full-width mask defines vessel pixels. The skeleton defines the graph-like centreline used for internal hemodynamics. The widthCalibrated map is used to estimate vessel radius where possible; otherwise, the Euclidean distance transform of the full-width vessel mask is used as a fallback radius estimate. The auto_domain_from_mat option sets the physical domain size from the image dimensions and MicronPerPixel calibration, thereby linking the simulation grid to the same physical calibration used in LeafVeinCNN and graph export.

3.2. Governing tissue oxygen equation

The tissue field is the dissolved oxygen concentration C(x,y,t), reported in mM and converted to pO2 in mmHg using the solubility coefficient alpha. The numerical tissue model follows the screening equation:

dC/dt = D laplacian(C) - Veff(C) C/(Km + C) + k_eff(x,y) [C_lumen(x,y) - C(x,y)].

Here D is the tissue oxygen diffusivity, Veff is the effective oxygen consumption rate, Km is the Michaelis-Menten half-saturation concentration, k_eff is the local vessel-to-tissue exchange coefficient, and C_lumen is the dissolved oxygen concentration assigned to perfused vessel pixels. Vessel pixels are excluded from tissue consumption by setting Vmax to zero inside the vessel mask. If the optional Pasteur term is enabled, Vmax is multiplied by f_pasteur(C) = [C/(C + K_pasteur)]^beta. This makes the consumption

term oxygen-dependent beyond the Michaelis-Menten factor, but it remains a modelling assumption rather than a measured property of the scaffold.

The tissue equation is solved on a two-dimensional Cartesian grid using an explicit finite-difference update. The interior update includes diffusion in x and y, Michaelis-Menten oxygen uptake, and vessel exchange. Boundary pixels use no-flux Neumann copying from adjacent interior pixels. Negative concentrations are clamped to zero after each update. A CFL guard reduces the time step when the explicit diffusion step would otherwise be unstable.

3.3. Radius-aware wall exchange

The exchange map is computed only on vessel pixels. The model uses the local approximation k(x,y) approximately equals 2 P_wall / r(x,y), where P_wall is the effective wall permeability or mass-transfer coefficient and r(x,y) is the local vessel radius. The radius comes from half of widthCalibrated when available or from the distance transform of the vessel mask when widthCalibrated is absent or unusable. A minimum radius floor prevents division by zero. A radius_scale multiplier is available for sensitivity analysis because image-derived vein width may not equal the true hydraulic lumen diameter.

The exchange coefficient is clamped so that dt times k does not exceed the exchange_clamp value. In the template configuration, exchange_clamp is 0.9. This is a numerical safeguard: it prevents the explicit update from removing more oxygen than is stable in one step. It must be reported as a stabilizing parameter, not as a biological measurement of plant wall permeability.

3.4. Internal skeleton hemodynamics

The solver constructs a hemodynamic graph directly from the vessel skeleton. Each skeleton pixel becomes a graph node connected to its four-neighbour skeleton pixels. Inlet and outlet candidates are selected from skeleton endpoints near image borders, and deterministic single-inlet/single-outlet fallbacks are available for disconnected or poorly terminated components. In the template configuration, single_io is true, which forces a simplified single-inlet/single-outlet interpretation. This improves numerical robustness but weakens biological realism if the actual scaffold would have multiple physical outlets.

For each adjacent skeleton-pixel pair, the local hydraulic conductance is computed from a Poiseuille-type expression, G = pi r^4/(8 mu ds), where r is the local vessel radius, mu is the fluid viscosity, and ds is the physical distance between neighbouring skeleton pixels. The pressure system A p = b is assembled as a sparse linear system. Boundary rows impose fixed inlet and outlet pressures, whereas interior rows enforce conductance-weighted conservation. After solving for node pressure, the pressure difference across each skeleton edge is converted into a local flow estimate and then into a mean nodal speed. A minimum

velocity floor u_min is applied to avoid division by zero and unbounded residence times in low-flow regions.

The code also computes reachability from the inlet by breadth-first search on the skeleton and extends the perfused label from reachable skeleton pixels to the full-width vessel components touched by those skeleton pixels. It then computes a transit-time map using Dijkstra shortest paths with edge weights ds/max(u,u_min). This map is labelled as RBC transit time in the code, but because the current template uses Hb_capacity = 0.0 and a blood-free viscosity value, the safer manuscript wording is convective transit-time proxy rather than true red-blood-cell transit time.

3.5. Lumen oxygen transport and haemoglobin option

The lumen module tracks total intraluminal oxygen along the skeleton. Total oxygen is defined as dissolved oxygen plus an optional haemoglobin-bound term: C_tot = C_diss + Hb_capacity S(pO2), where C_diss = alpha pO2 and S(pO2) is a Hill saturation function. The solver includes bisection routines to convert between total oxygen and pO2, and it performs a round-trip self-test at startup. In the present configuration, Hb_capacity is set to 0.0, so the haemoglobin reservoir is disabled and the lumen model reduces to dissolved oxygen transport.

Lumen oxygen is propagated along the skeleton using geodesic distance from the inlet. For each reachable skeleton pixel, the algorithm selects the neighbouring skeleton pixel with the smallest geodesic distance as the upstream source. The local oxygen loss is computed with lambda = k ds/u, where k is the local exchange coefficient, ds is the grid step, and u is the local speed after applying the velocity floor. The code then updates total oxygen as C_tot,new = C_tot,up - lambda max(C_diss,up - C_tissue, 0), with lambda capped by exchange_clamp and the result clipped between zero and the inlet total oxygen. After this skeleton oxygen calculation, dissolved lumen oxygen is lifted to all perfused vessel pixels using a nearest-reachable-skeleton mapping.

This design avoids the dense Green-function matrix problem that stopped GreensV4. However, it does not fully solve cyclic intravascular solute transport. It imposes a geodesic upstream direction based on distance from the inlet and therefore approximates loops rather than solving a simultaneous loop-mixing system. This is one of the main limitations of the proposed solver and must be stated clearly.

3.6. Coupled simulation loop

The simulation begins with zero tissue oxygen and computes an initial lumen field from the hemodynamic cache. During the time loop, lumen oxygen is rebuilt only every rebuild_every steps and only when the near-vessel tissue oxygen band changes by more than rebuild_tol. This reduces computational cost because recalculating the lumen map is

expensive relative to one tissue finite-difference update. The tissue field is then advanced with diffusion, consumption, and exchange. Intermediate tissue fields are saved every step interval specified in the configuration.

Convergence is monitored by a relative L2 error between the current and previous check fields. The code can check the whole tissue grid or only the near-vessel band. The simulation stops early if stop_on_converge is true and the relative or optional absolute tolerance is satisfied after the required minimum number of checks. A maximum wall-clock runtime can also stop the run. These controls make the solver practical for parameter sweeps, but they also mean that every reported result must include the convergence and runtime settings used.

3.7. Exactly what the solver calculates

The solver calculates four classes of outputs. First, it calculates geometry-derived maps: vessel mask, skeleton, radius map, perfused-vessel mask, and distance-to-nearest-vessel distributions. Second, it calculates internal hemodynamic descriptors on the skeleton: pressure field, speed map, reachability, and convective transit-time proxy. Third, it calculates oxygen fields: final tissue dissolved oxygen in mM, final tissue pO2 in mmHg, lumen dissolved oxygen in mM, and lumen pO2 in mmHg. Fourth, it calculates scalar screening metrics: minimum, median, mean, percentile, and maximum tissue oxygen; lumen oxygen ranges; tissue area fractions at or above 15, 20, and 30 mmHg; fractions of tissue within 50, 100, and 200 μm of a vessel; and optional timing diagnostics such as diffusion-time estimates.

The solver also writes figures and machine-readable files. The main numerical arrays are saved as .npy files for exact reanalysis: tissue_O2_mM.npy, tissue_pO2_mmHg.npy, lumen_O2_mM.npy, lumen_pO2_mmHg.npy, and vessel_mask.npy. Summary outputs are written as oxygenation_summary.csv and oxygenation_summary.json. Additional metrics are written to metrics.txt, metrics.json, and metrics.csv. The reported area-above-threshold values should be interpreted as oxygenation screen metrics, not as direct measurements of cell viability.

Algorithmic structure of the proposed OxySim oxygen-screening solver.

Step	**Code operation**	**Quantity calculated**	**Scientific interpretation**
1	Read JSON config and LeafVeinCNN MAT export.	Domain size, mask, skeleton, width map, calibration.	Defines reproducible physical geometry and parameter set.
2	Resize mask/skeleton to simulation grid.	vessel_mask and skeleton_grid.	Places image-derived scaffold geometry onto numerical grid.

Step	Code operation	Quantity calculated	Scientific interpretation
3	Compute radius-aware exchange.	r(x,y) and k(x,y) = 2 P_wall/r.	Approximates vessel-wall transfer; uncertain and swept.
4	Solve skeleton Poiseuille system.	Pressure, speed, reachability, perfused mask.	Internal perfusion proxy, not direct NetFlowV2 output.
5	Propagate lumen oxygen.	C_lumen on skeleton and vessel mask.	Approximate convective supply with exchange loss.
6	Update tissue field explicitly.	C_tissue(x,y,t).	2-D reaction-diffusion-consumption oxygen screen.
7	Check convergence and save outputs.	O2 arrays, pO2 maps, threshold fractions, CDFs.	Produces reproducible screening metrics for comparison.

3.8. Batch sweep execution

Parameter sweeps were automated using run_o2_sweep_fast.py. This script reads a base JSON file, generates run-specific configuration files, launches oxy_sim_mat_saveO2.py as a subprocess, saves stdout and stderr logs, and collects successful oxygenation summaries into MASTER_oxygenation_summary.csv. It also records failed runs in MASTER_failed_runs.csv. The sweep dimensions are P_wall, Vmax, and inlet_mmHg. In the default full mode, the reduced design uses P_wall values 1e-8, 1e-6, 3e-5, and 1e-4 m/s; Vmax values 1e-5, 1e-4, and 1e-3 mM/s; and inlet pO2 values 80 and 150 mmHg. This produces 24 parameter combinations.

The batch runner is important methodologically because a single oxygen map is not meaningful when the key biological parameters are uncertain. The correct interpretation is comparative and conditional: under a specified P_wall, Vmax, and inlet oxygen condition, the model reports how much of the tissue grid exceeds operational oxygen thresholds. It does not infer the true plant-wall permeability or the true metabolism of seeded cells.

Base configuration values used by the MAT-direct OxySim template.

Parameter group	Key values	Role in solver	Claim limit
Grid/time	nx = 384, ny = 384, T = 300 s, nt = 1000	Defines numerical grid and simulation duration.	Resolution-dependent; must pass stability and convergence checks.

Parameter group	Key values	Role in solver	Claim limit
Diffusion/consumption	D = 1.5e-9 m2/s, Vmax = 1e-4 mM/s, Km = 0.001 mM	Controls tissue oxygen spread and uptake.	Not scaffold-specific measurements.
Exchange	P_wall = 3e-5 m/s, exchange_clamp = 0.9, radius_scale = 1.0	Controls vessel-to-tissue oxygen transfer.	P_wall is the dominant uncertain parameter.
Inlet oxygen	inlet_mmHg = 150, alpha = 0.00134 mM/mmHg	Sets dissolved oxygen entering vessel network.	Depends on medium, temperature, and gas equilibration.
Hemodynamics	P_in = 40 mmHg, P_out = 39 mmHg, mu = 0.0007 Pa.s, u_min = 1e-4 m/s	Builds internal pressure/speed proxy.	Not a measured scaffold perfusion experiment.
Haemoglobin	Hb_capacity = 0.0, P50 = 26 mmHg, Hill n = 2.7	Hb module present but disabled in current run.	Do not interpret as blood unless Hb_capacity is nonzero and justified.
Stopping	rebuild_every = 25, rebuild_tol = 0.01, tol = 1e-6, max_minutes = 20	Controls lumen rebuild and convergence.	Must be reported because outputs can depend on convergence behavior.

3.9. Limitations of the proposed solver

The solver reconstructs hemodynamics internally from the resized LeafVeinCNN skeleton; it does not directly consume NetFlowV2 segment flows. Therefore, it is a proposed oxygen-screening solver, not a validated NetFlowV2 downstream oxygen solver.

The tissue domain is two-dimensional. It does not resolve scaffold thickness, three-dimensional cell distribution, wall microstructure, matrix tortuosity, or heterogeneous seeded-cell density.

The lumen transport step uses geodesic ordering from the inlet and a single upstream neighbour. This is computationally robust but does not fully solve cyclic loop mixing in reticulated networks.

P_wall, Vmax, Km, and the effective hydraulic radius are sensitivity parameters rather than measured properties of the processed scaffold.

The velocity floor u_min and exchange_clamp are numerical safeguards. They prevent division-by-zero and over-drainage but may artificially influence low-flow regions.

The reported viable-area or cell-slot metrics are threshold-based oxygenation summaries. They are not experimental viability measurements and should be renamed oxygen-threshold area fractions in the main claims.

The current Hb_capacity is zero in the template, so haemoglobin buffering is disabled. Any future Hb-enabled run would need a separate biological justification and a different interpretation of the perfusate.

Resizing the mask and skeleton to the simulation grid can alter very fine topology. The same grid size and calibration must be used consistently across runs when comparing images.

3.10. Equation justification and source discipline

The equations used in the proposed OxySim solver were selected as a screening-level combination of established transport components, not as a new biological law. Tissue oxygen movement was represented by Fickian diffusion with a reaction term, which is the standard structure used in microvascular and tissue-engineering oxygen models. Oxygen consumption was represented using Michaelis-Menten kinetics because this form is widely used for oxygen uptake in engineered tissues and because measured three-dimensional constructs show that Vmax and Km are condition-specific parameters rather than universal constants (Michaelis and Menten, 1913; Casciari et al., 1992; Magliaro et al., 2019).

The Poiseuille-type conductance equation was justified only for the internal skeleton-hemodynamics approximation. It expresses the expected fourth-power dependence of hydraulic conductance on radius in laminar tube flow and is consistent with the pressure-flow formulation used in microvascular network simulations (Pries et al., 1990; Pries and Secomb, 2008). In this project it should not be interpreted as proof that the segmented vein width equals the true open hydraulic lumen.

The radius-aware exchange coefficient k approximately equal to 2 P_wall/r was introduced as a lumped conversion from a cylindrical wall flux into a volumetric source term. The factor 2/r is the surface-area-to-volume ratio of a cylinder, while P_wall is treated as an effective permeability or mass-transfer coefficient. This is consistent with the way microvascular oxygen models represent transvascular exchange as a permeability-limited flux, but P_wall is not measured for the present decellularized scaffold and must therefore be swept rather than fixed (Hsu and Secomb, 1989; Secomb et al., 2004).

The optional haemoglobin module follows the Hill saturation form for oxygen binding, with total oxygen written as dissolved oxygen plus a capacity-weighted saturation term. This component is included for code generality and for possible oxygen-carrier simulations. However, in the present template Hb_capacity is 0.0, so haemoglobin buffering is disabled. The manuscript must therefore avoid blood-specific claims unless a separate Hb-enabled parameterization is justified biologically (Hill, 1910; Pries and Secomb, 2005).

The distance-to-vessel and diffusion-time metrics are justified as conservative geometry and

timing descriptors. Krogh-style oxygen calculations established the importance of capillary spacing and diffusion distance, and later microvascular oxygen models emphasized that tissue oxygenation depends strongly on microvessel spatial arrangement. These metrics are therefore defensible as scaffold-accessibility descriptors, but they are not measurements of cell survival (Krogh, 1919; Secomb et al., 2004; Magliaro et al., 2019).

Table 2. Source justification for the proposed OxySim equations and numerical terms.

Model term	**Equation or operation**	**Source justification**	**Claim limit in this project**
Tissue diffusion-reaction	$\partial C/\partial t = D\nabla^2 C - V_{\text{eff}} C/(K_m + C) + k_{\text{eff}}(C_{\text{lumen}} - C)$	Fickian diffusion with reaction is standard in oxygen transport models; Michaelis-Menten oxygen uptake is used in tissue constructs and oxygen-consuming cell systems (Michaelis and Menten, 1913; Casciari et al., 1992; Magliaro et al., 2019).	Screening-level tissue oxygen field only; not experimentally calibrated scaffold physiology.
Poiseuille skeleton hemodynamics	$G = \pi r^4/(8\mu ds)$; $Ap = b$	Microvascular network flow models use pressure-flow relations with conductance depending strongly on vessel diameter (Pries et al., 1990; Pries and Secomb, 2008).	Internal pixel-skeleton estimate; not a replacement for NetFlowV2 and not validated plant perfusion.
Radius-aware exchange	$k \approx 2P_{\text{wall}}/r$	Derived from cylindrical surface-area-to-volume scaling and permeability-limited wall exchange used in vessel-to-tissue oxygen models (Hsu and Secomb, 1989; Secomb et al., 2004).	P_wall is unknown for the scaffold; sweep as sensitivity parameter.

Model term	Equation or operation	Source justification	Claim limit in this project
Lumen oxygen loss	lambda = k ds/u; Ctot,new = Ctot,up - lambda max(Cdiss,up - Ctissue,0)	Represents convective residence-time-dependent loss from lumen to tissue; conceptually consistent with intravascular transport plus transvascular exchange in microvascular oxygen models (Secomb et al., 2004).	Uses geodesic upstream approximation; does not solve full loop mixing.
Hb option	Ctot = Cdiss + Hb_capacity S(pO2); S = pˆn/(P50ˆn + pˆn)	Hill saturation is the classical empirical model for haemoglobin oxygen dissociation (Hill, 1910).	Disabled in current template because Hb_capacity = 0.0; no blood-carrier claim.
Dissolved O2 and pO2 conversion	Cdiss = alpha pO2	Henry-type solubility conversion is used in oxygen transport calculations; oxygen solubility depends on medium and temperature (Christoforides et al., 1969; Goldstick and Fatt, 1970).	alpha is an assumed conversion coefficient, not measured in the scaffold medium.
Distance and diffusion timing	tdiff = dˆ2/(2D)	Krogh-style oxygen calculations and later microvascular models link oxygenation to vessel spacing and diffusion distance (Krogh, 1919; Secomb et al., 2004).	Geometry/timing descriptor only; not a viability measurement.

Model term	Equation or operation	Source justification	Claim limit in this project
Explicit numerical update	CFL guard; no-flux boundary copy; non-negativity clamp	Finite-difference stability constraints are required for explicit diffusion schemes (Strikwerda, 2004).	Numerical safeguard; must be reported with grid and time-step settings.

3.11. Parameterization and sweep design

The final workflow retained parameter sweeps instead of single-value biological claims. This was necessary because wall permeability, effective seeded-cell oxygen demand, and plant-scaffold transport properties are not directly calibrated for the available images.

Parameter sweep used for the oxygen-screening stage.

Parameter	Values	Interpretation
P_wall	1×10^{-6}, 3×10^{-5}, 1×10^{-4} m/s	Low, baseline, and high effective wall-exchange regimes.
Vmax	1×10^{-5}, 1×10^{-4}, 1×10^{-3} mM/s	Sparse, baseline, and dense seeded-cell oxygen-demand regimes.
Inlet oxygen	150 mmHg; approximately 0.20 mM dissolved O2	Air-equilibrated aqueous perfusate approximation.
Tissue thresholds	15, 20, and 30 mmHg	Operational oxygenation thresholds for comparing simulated conditions.

3.12. Verification boundary and R Markdown stability analysis

A separate R Markdown validation document is used to analyze the numerical behavior of the proposed oxygen-screening solver. Its role is deliberately restricted. It is used to evaluate stability, reproducibility, and directional consistency under the selected parameter sweep; it is not used as proof that the simulated scaffold is biologically correct.

The validation logic separates three claims. First, computational executability means that the code produces finite arrays and machine-readable summaries without runtime collapse. Second, numerical stability means that oxygen fields remain bounded, non-negative after clamping, and directionally consistent when wall exchange, consumption, and inlet oxygen are varied. Third, biological validity would require independent experimental oxygen measurements or calibrated scaffold permeability data. Only the first two claims are supported in the present project.

This distinction is essential because decellularized plant scaffolds have uncertain wall permeability, leakage, lumen continuity, and recellularization behavior. The absence of a wet-lab oxygen reference prevents a definitive viability claim.

4. Results

4.1. Image-to-flow conversion was computationally feasible

The upstream workflow successfully produced NetFlowV2-readable graph files from image-derived vascular data. This result is important because it establishes that the plant scaffold can be represented as a computational vascular graph with explicit nodes, segments, radii, and boundary nodes.

The successful output of NetFlowV2 should not be interpreted as a complete oxygenation result. It establishes the hemodynamic state of the graph; oxygen delivery still requires a downstream transport model.

Demonstration NetFlow-derived graph metrics used for the oxygen-screening prototype.

Metric	**Value**
Total segments in demonstration network	1130
Active nonzero-flow segments	1130
Total graph nodes	972
Boundary nodes	36
Detected inlet nodes	31
Detected outlet nodes	5
Boundary inflow	776.162 nL/min
Boundary outflow	776.162 nL/min
Boundary balance error	3.000e-06 nL/min

5. Discussion

5.1. Scientific meaning of the negative result

The GreensV4 failure is scientifically useful because it identifies a domain-of-validity problem. A solver developed for microvascular oxygen delivery cannot be assumed to work on reticulated plant networks simply because both systems contain vessel-like graphs. The pressure-flow problem and the intravascular oxygen-transport problem impose different computational requirements.

The main methodological lesson is that algorithmic assumptions must match biological

topology. Leaf venation is explicitly reticulated. Forcing such a graph into a tree-like downstream ordering by pruning loops would remove the property that makes plant venation attractive as a scaffold: redundant transport architecture.

5.2. Why NetFlowV2 remains useful

NetFlowV2 remains appropriate for the hemodynamic stage because it solves a global pressure-flow problem on the network. The observed GreensV4 failure does not invalidate the NetFlow output. Instead, it indicates that the oxygen stage must be implemented using a method compatible with cyclic directed dependencies.

5.3. Status of the proposed oxygen-screening solver

The proposed oxygen-screening solver is a methodologically defensible contribution, provided that its claim boundary is restricted. It addresses the specific failure encountered in GreensV4 by using sparse pixel-skeleton hemodynamics, radius-aware wall exchange, and explicit two-dimensional reaction-diffusion tissue updating.

The most defensible interpretation is therefore methodological: the project produced a reproducible image-to-flow pipeline, identified why a reference Green-function solver failed on reticulated plant venation, and implemented a preliminary solver whose behavior can be screened across uncertain parameters. Publication-level biological claims would require independent calibration of plant-wall permeability, seeded-cell oxygen consumption, perfusion leakage, and oxygen concentration measurements.

6. Limitations

The proposed oxygen-screening solver is not yet a validated physiological oxygen transport model. It reconstructs pixel-skeleton hemodynamics internally, uses a two-dimensional tissue grid, applies numerical safeguards such as a velocity floor and exchange clamp, and approximates looped lumen transport through geodesic ordering. These choices make the solver executable and auditable, but they also prevent it from being treated as a definitive physiological model.

Consequently, OxySim outputs should be reported as oxygen-screening metrics: tissue pO2 distributions, threshold area fractions, lumen oxygen maps, distance-to-vessel summaries, and sensitivity trends. They should not be reported as proof that the scaffold sustains mammalian cell viability.

The pixel-to-micrometre calibration used in LeafVeinCNN was fixed at 9.62 μm/px for the processed images and was propagated consistently through MAT export, graph conversion, and oxygen screening. No wet-lab oxygen measurements were available; therefore, biological calibration and validation were not possible.

7. Conclusion

This project developed a reproducible computational workflow for converting image-derived leaf venation into NetFlowV2-compatible vascular graphs and hemodynamic outputs. The initial objective was to couple the extracted plant vascular geometry to GreensV4 for downstream oxygen-transport modelling. This approach failed for specific and interpretable reasons: the dense Green-function interaction matrices exceeded practical memory limits at full image-derived resolution, and the intravascular convective transport routine relied on a valid downstream ordering that can be violated by reticulated plant venation. GreensV4 was therefore not treated as a failed implementation, but as a solver operating outside its reliable domain for the tested full-resolution reticulated scaffold geometry.

The proposed oxygen-screening solver was introduced as a controlled computational alternative. Its role is limited to preliminary oxygen-accessibility screening under explicit numerical and biological assumptions. Its stability is evaluated separately through the R Markdown validation workflow, and its outputs are not presented as experimentally validated predictions of scaffold viability.

The final contribution of this capstone is a rigorous negative result combined with a documented computational alternative. The completed workflow, GreensV4 failure analysis, parameter sweep, and stability audit support a reproducible screening framework for plant-derived scaffold oxygenation. However, experimental calibration remains necessary before any claim can be made about true biological viability, tissue survival, or clinical relevance.

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